



$$\begin{cases} \partial_t f = \sigma_x \Delta_x f - \text{Pe} \nabla_x \cdot (v(\theta) f) + \sigma_\theta \partial_\theta^2 f - \chi \partial_\theta (n(\theta) \cdot (\nabla K * \rho + \tau \nabla^2 K * \rho v(\theta)) f) \\ f(t=0) = f_0 \end{cases}$$

- What happens as  $t \rightarrow \infty$ ?

• Part 1 (Rakotomalala & W, in review, 2025):

o Existence of compact global attractor

■ Robinson, *Infinite-Dimensional Systems*, 2001

◦ For small enough  $\sigma_x, \sigma_\theta$ , the attractor has dimensional lower bound of 4 if homogenous state is linearly unstable



# Lypunov-Peron method





**Dynamics**

# Dynamics

$$\begin{cases} \partial_t f &= \sigma_x \Delta_x f - \text{Pe} \nabla_x \cdot (v(\theta) f) + \sigma_\theta \partial_\theta^2 f - \chi \partial_\theta (n(\theta) \cdot (\nabla K * \rho + \tau \nabla^2 K * \rho v(\theta)) f) \\ f(t=0) &= f_0 \end{cases}$$

- What happens as  $t \rightarrow \infty$ ?
- Part 1 (Rakotomalala & dW, in review, 2025):
  - Existence of compact global attractor
    - Robinson, *Infinite-Dimensional Dynamical Systems*, 2001
  - For small enough  $\sigma_x, \sigma_\theta$ , the attractor has dimensional lower bound of 4 if homogeneous state is linearly unstable
    - Lyapunov-Perron method



# Dynamics

$$\begin{cases} \partial_t f &= \sigma_x \Delta_x f - \text{Pe} \nabla_x \cdot (v(\theta) f) + \sigma_\theta \partial_\theta^2 f - \chi \partial_\theta (n(\theta) \cdot (\nabla K * \rho + \tau \nabla^2 K * \rho v(\theta)) f) \\ f(t=0) &= f_0 \end{cases}$$

- What happens as  $t \rightarrow \infty$ ?
- Part 2 (Rakotomalala & dW, in review, 2026):

